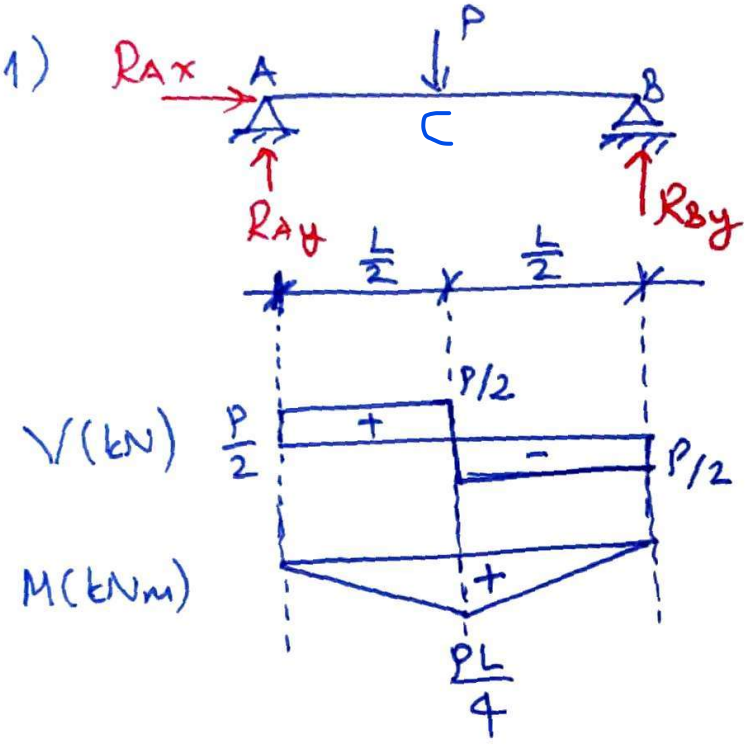
# INTERNAL FORCE DIAGRAMS of SIMPLE SYSTEMS
## (Simply Supported Beams - Cantilever Beams)

1)

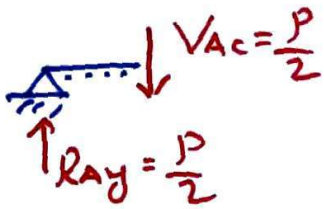



**Support Reactions:**

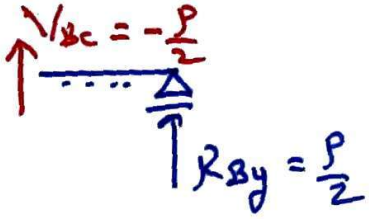
$\Sigma P_x = 0 \Rightarrow \quad R_{Ax} = 0$

$\Sigma P_y = 0 \Rightarrow \quad R_{Ay} = R_{By} = \frac{P}{2}$

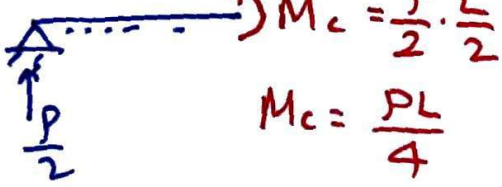
$V_{Ac} \Rightarrow$



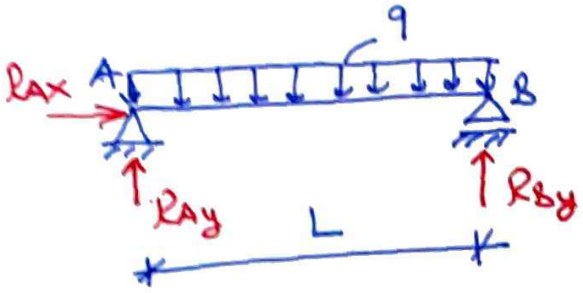

$V_{Bc} \Rightarrow$




$M_c \Rightarrow$



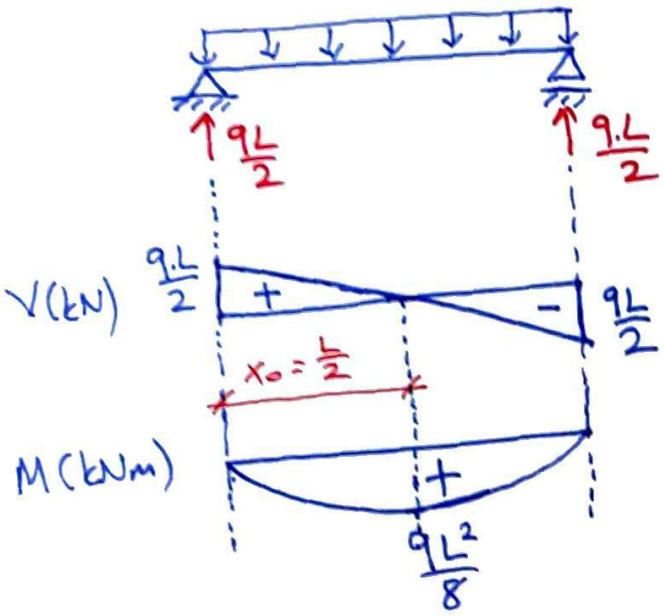
2)




Support Reactions:

$\Sigma F_x = 0 \Rightarrow R_{Ax} = 0$

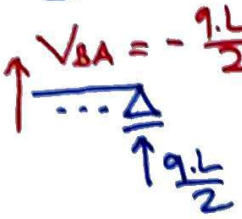
$\Sigma F_y = 0 \Rightarrow R_{Ay} = R_{By} = \frac{q.L}{2}$

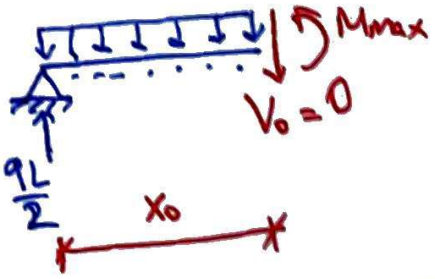



$V_{AB} \Rightarrow$




$V_{BA} \Rightarrow$



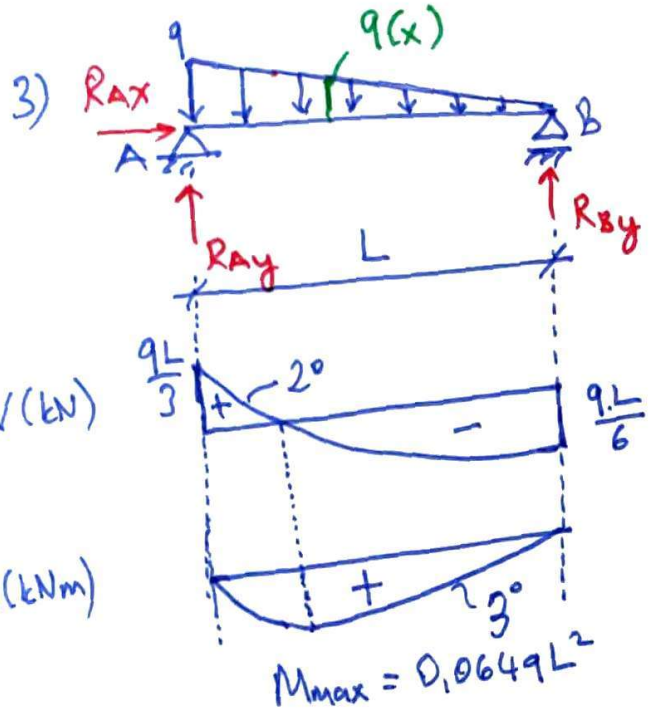




$q.x_0 = \frac{q.L}{2} \Rightarrow x_0 = \frac{L}{2}$

$M_{max} = \frac{q.L}{2}.\frac{L}{2} - \frac{q.L}{2}.\frac{L}{4}$

$= \frac{q.L^2}{4} - \frac{q.L^2}{8}$

$M_{max} = \frac{q.L^2}{8}$

3) 


$M_{max} = 0{,}0649L^2$

## Support Reactions:

$\Sigma F_x = 0 \Rightarrow R_{Ax} = 0$

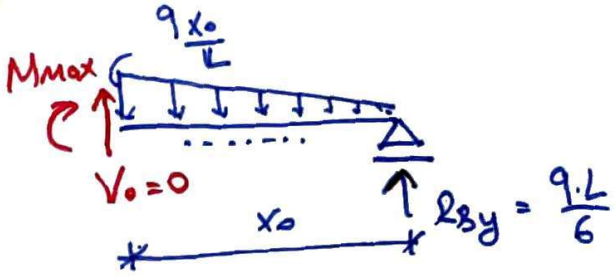
$\Sigma M_B = 0$

$R_{Ay} = 9.L\frac{1}{2}.\frac{2L}{3}\frac{1}{L} = \frac{9.L}{3}$

$\Sigma M_A = 0$

$R_{By} = 9.L.\frac{1}{2}.\frac{L}{3}.\frac{1}{L} = \frac{9.L}{6}$

$q(x) = \frac{9x_o}{L}$




$V_o = 0 \Rightarrow \frac{9.L}{6} = \frac{9x_o}{L}.\frac{x_o}{2} \Rightarrow x_o = \frac{L}{\sqrt{3}}$

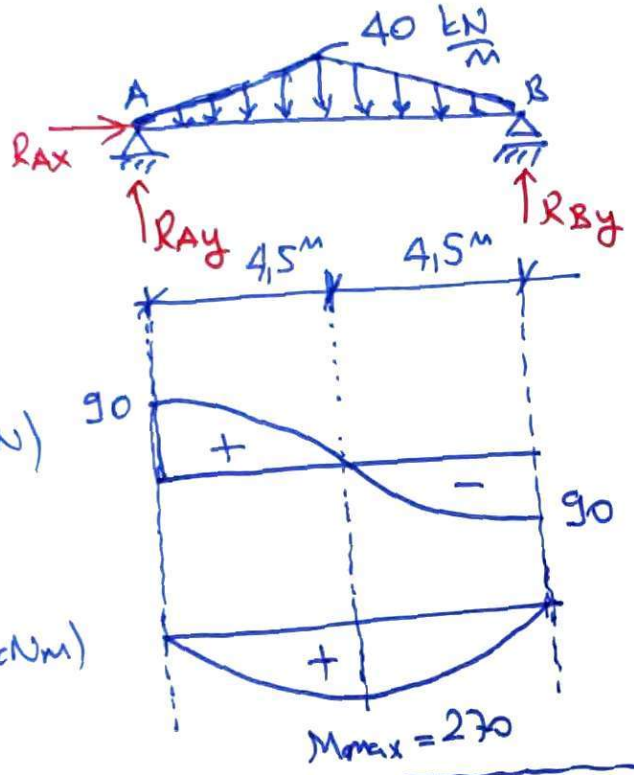
$M_{max} = \frac{9.L}{6}\frac{L}{\sqrt{3}} - \frac{9.}{\sqrt{3}}.\frac{L}{\sqrt{3}}\frac{1}{2}.\frac{L}{3\sqrt{3}} = \frac{9L^2}{6\sqrt{3}} - \frac{9L^2}{18\sqrt{3}}$

$M_{max} = 0{,}064.9L^2$

Second Way: Estimation of $M_{max}$ by using area of shear force diagram

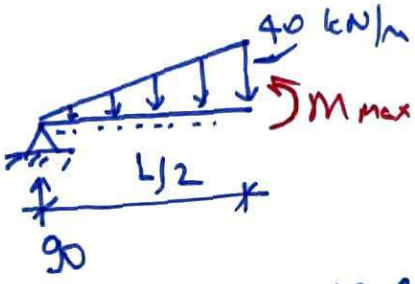
$M_{max} = \frac{2}{3}.\frac{9L}{6}.\frac{L}{\sqrt{3}} = 0{,}0649L^2$

4)




$\Sigma P_x = 0 \Rightarrow R_{Ax} = 0$

$\Sigma P_y = 0 \Rightarrow R_{Ay} = R_{By} = \frac{40 \cdot 9}{2} \frac{1}{2} = 90\,kN$

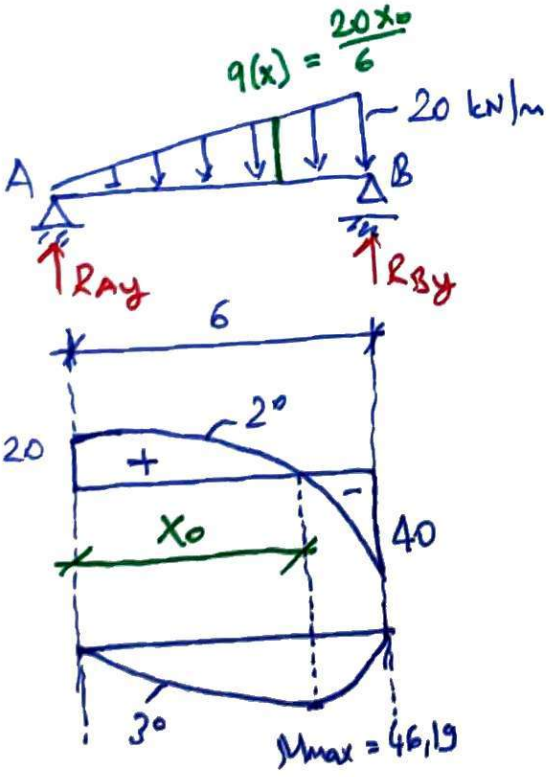



$M_{max} = 90 \cdot 4{,}5 - 40 \cdot \frac{4{,}5}{2} \cdot \frac{4{,}5}{3}$

$M_{max} = 270\,kNm$

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5)




$\Sigma M_B = 0$

$R_{Ay} = \frac{20 \cdot 6}{2} \cdot \frac{6}{3} \cdot \frac{1}{6} = 20\,kN$

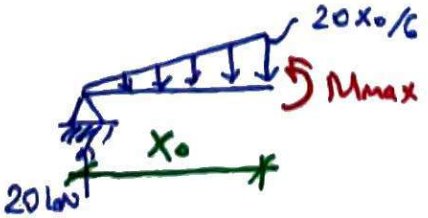
$\Sigma M_A = 0$

$R_{By} = \frac{20 \cdot 6}{2} \cdot \frac{2}{3} \cdot 6 \frac{1}{6} = 40\,kN$

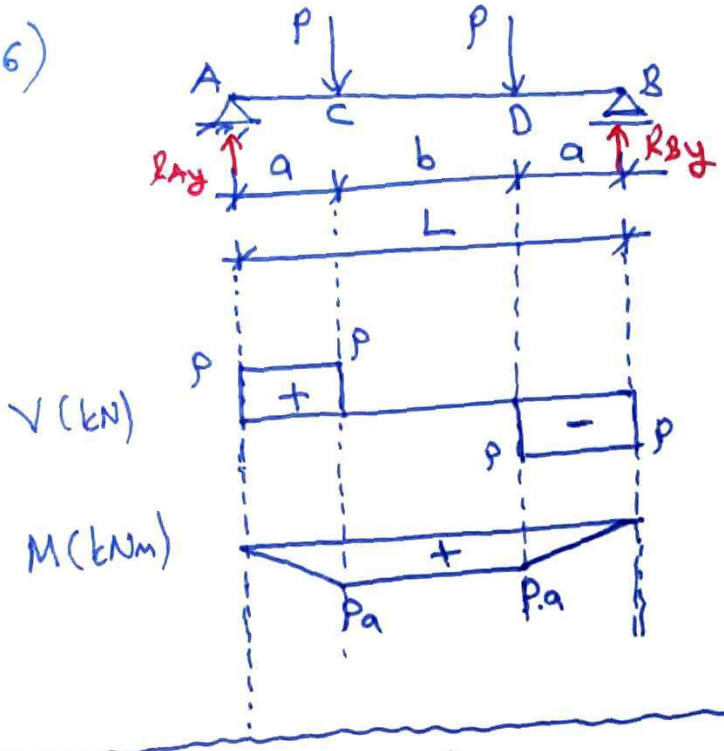
$20 = \frac{20 x_0}{6} \cdot x_0 \cdot \frac{1}{2} \Rightarrow x_0 = 3{,}46\,m$

$M_{max} = 20 \cdot 3{,}46 - 20 \cdot \frac{3{,}46}{6} \cdot \frac{3{,}46}{2} \cdot \frac{3{,}46}{3}$

$M_{max} = 46{,}19\,kNm$

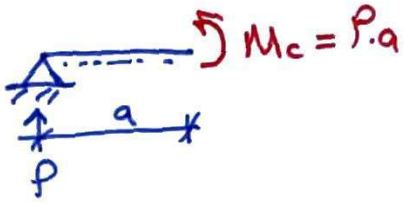


6)



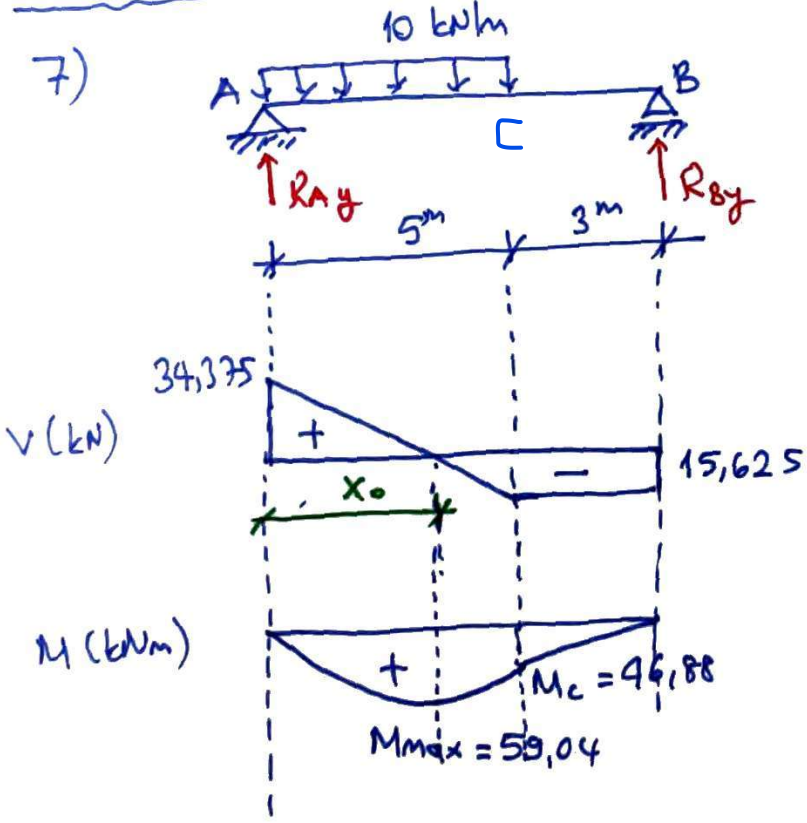

$$R_{Ay} = R_{By} = P$$

$$V_{AC} = -V_{BD} = P$$




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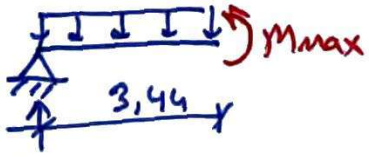
7)




$$R_{Ay} = \frac{10.5.5,5}{8} = 34,375\ kN$$

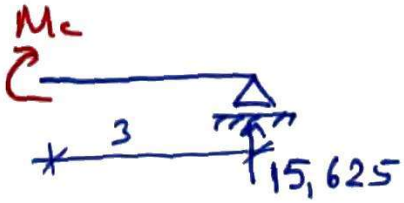
$$R_{By} = \frac{10.5.2,5}{8} = 15,625\ kN$$

$$x_0 = \frac{R_{Ay}}{q} = 3,44\ m$$



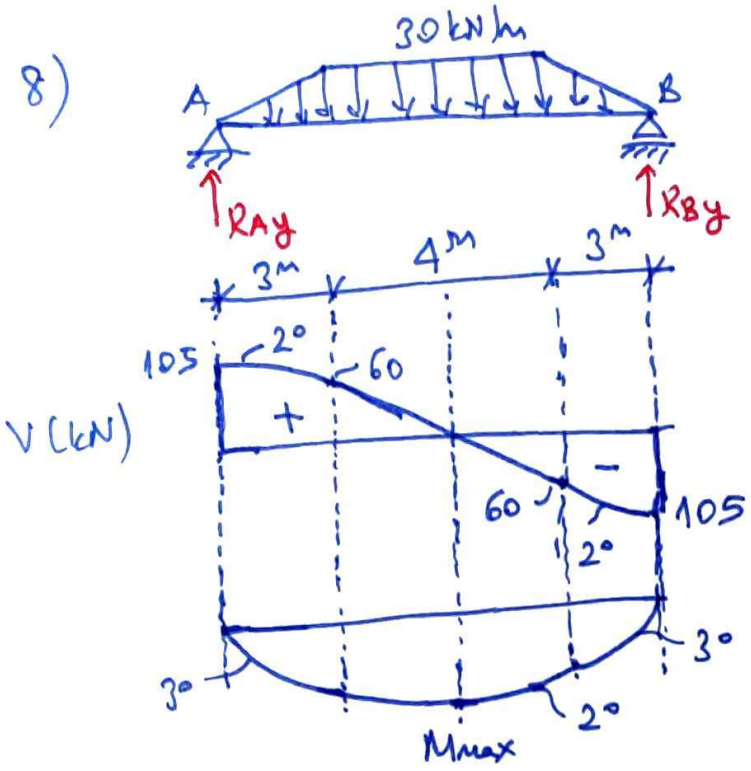

$$M_{max} = 34,375.3,44 - 10.\frac{3,44^2}{2}$$

$$M_{max} = 59,04\ kNm$$



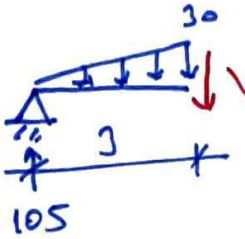

$$M_c = 15,625.3 = 46,88\ kNm$$

8)




$$R_{Ay} = R_{By} = \left[30.4 + 2.\left(\frac{30.3}{2}\right)\right]\frac{1}{2}$$

$$R_{Ay} = R_{By} = 105\ kN$$



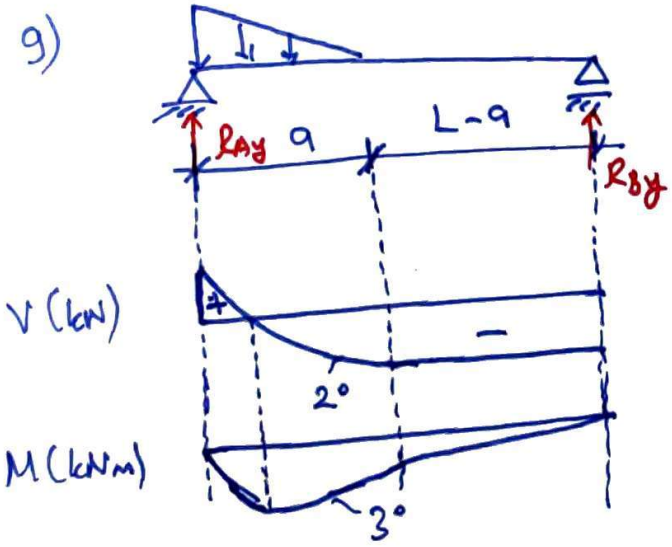

$$V_{CA} = 105 - \frac{30.3}{2} = 60\ kN$$

$$M_{max} = 105.5 - \frac{30.3}{2}.\left(2+\frac{3}{3}\right) - 30.2.1$$

$$M_{max} = 330\ kNm$$

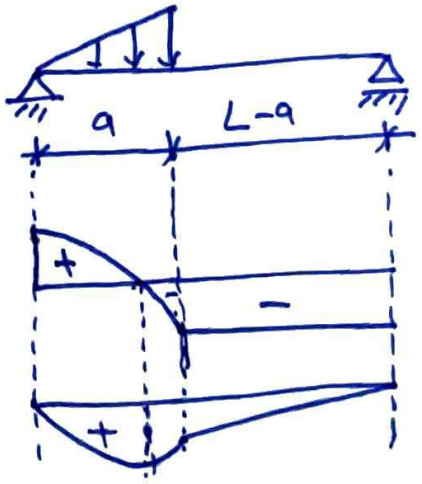
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9)




$$R_{Ay} = \frac{q.a}{2L}\left(L - \frac{a}{3}\right)$$

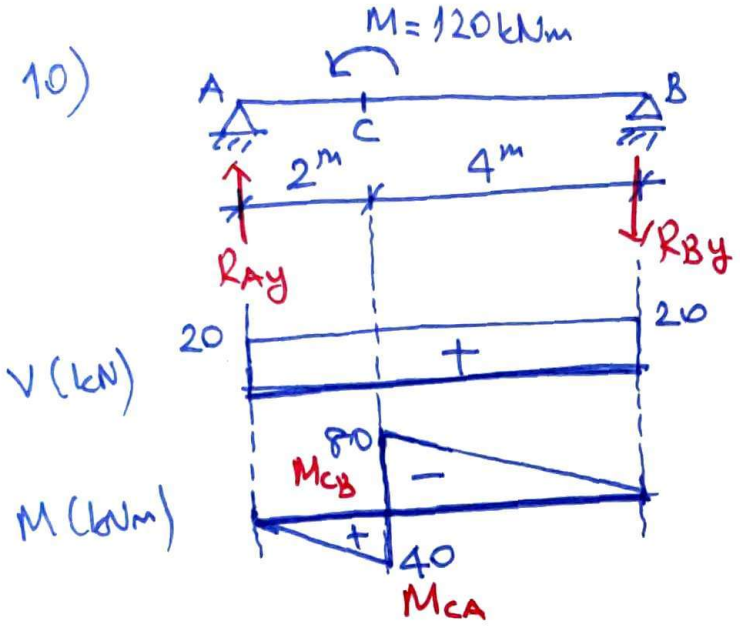
$$R_{By} = \frac{q.a^2}{6L}$$



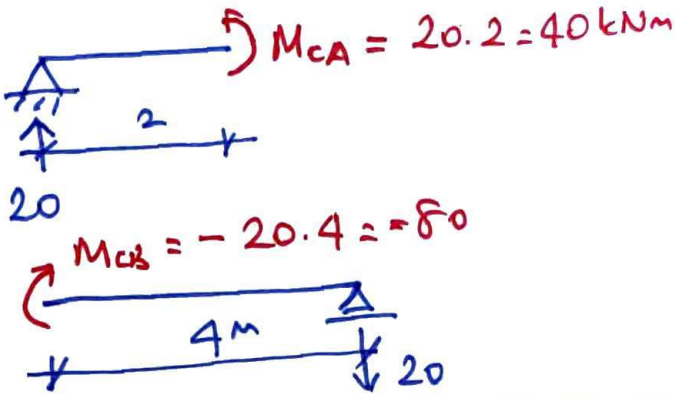

$$R_{Ay} = \frac{q.a}{2L}.\left(L - \frac{2}{3}a\right)$$

$$R_{By} = \frac{q.a^2}{3L}$$

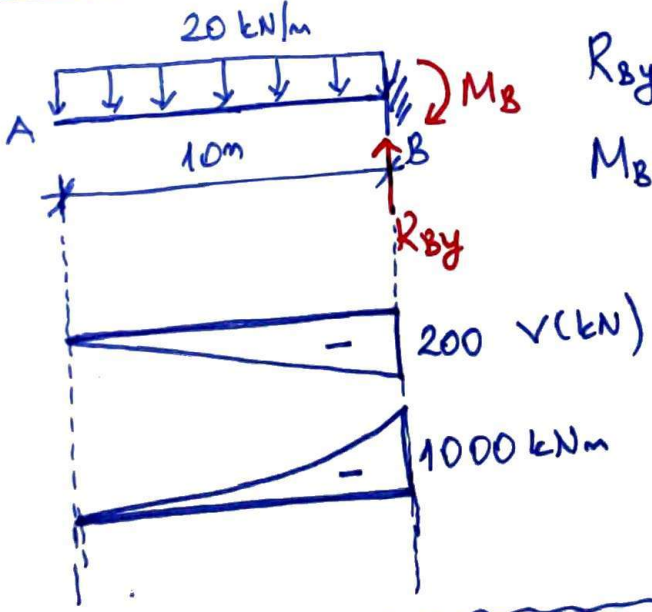
10)




$$R_{Ay} = +R_{By} = M/L = \frac{120}{6} = 20\ kN$$



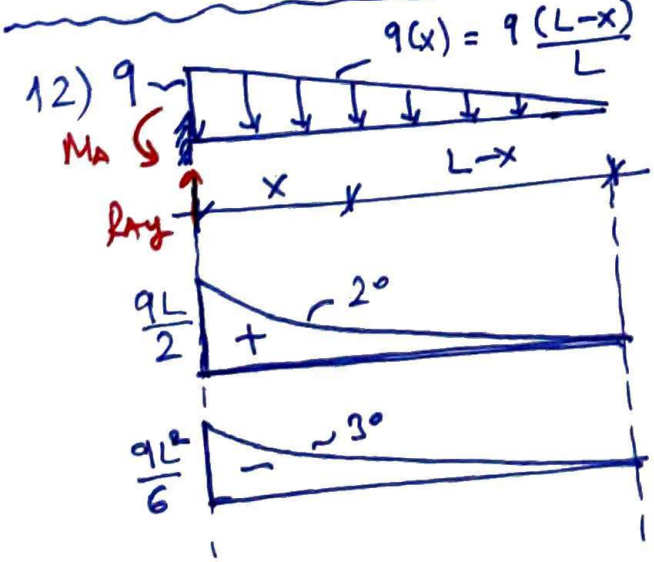

11)




$$R_{By} = 20.10 = 200\ kN$$

$$M_B = 20.10.5 = 1000\ kNm$$

12)



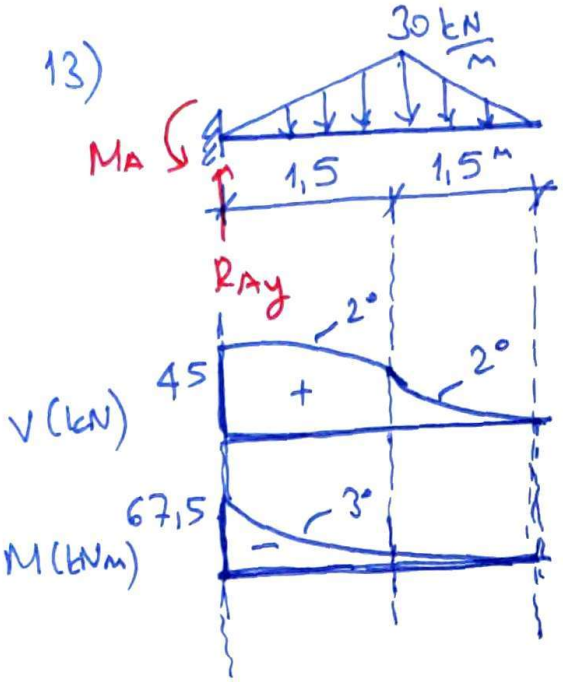

$$R_{Ay} = \frac{9L}{2}$$

$$M_A = \frac{9L}{2} \cdot \frac{L}{3} = \frac{9L^2}{6}$$

$$V(x) = \frac{9(L-x)}{L} \cdot \frac{(L-x)}{2}$$

$$M(x) = V(x)\ \frac{(L-x)}{3}$$

13)

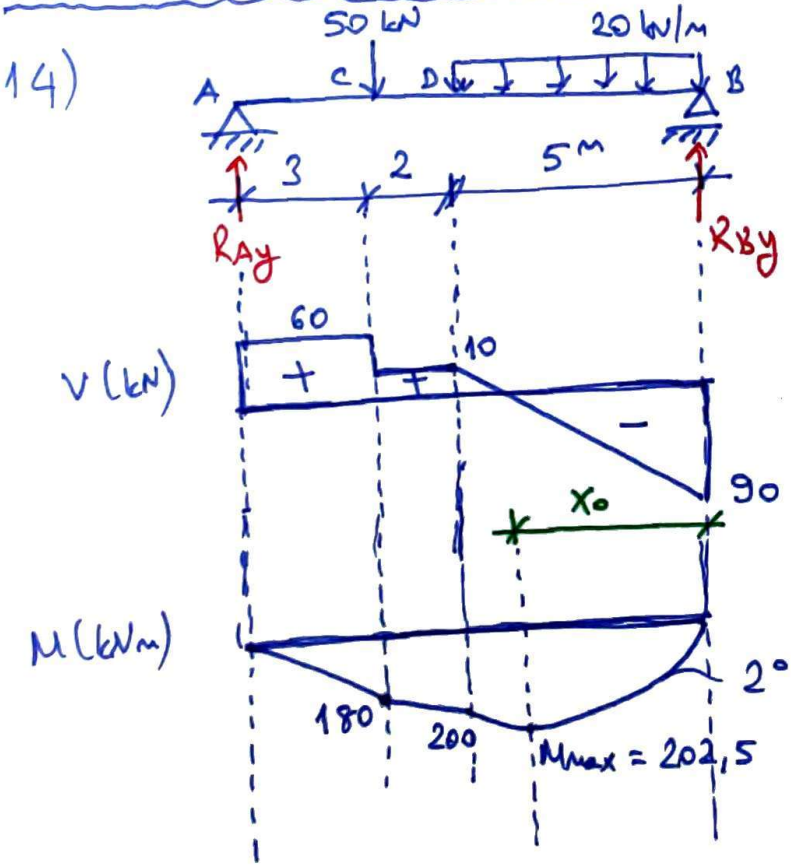



$$R_{Ay} = \frac{30.3}{2} = 45\,kN$$

$$M_A = \frac{30.3}{2}.1,5 = 67,5\,kNm$$

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14)

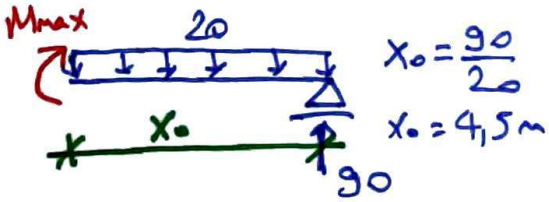



$$R_{Ay} = \frac{50.7 + 20.5.2,5}{10}$$

$$R_{Ay} = 60\,kN$$

$$R_{By} = 50 + 20.5 - 60$$

$$R_{By} = 90\,kN$$



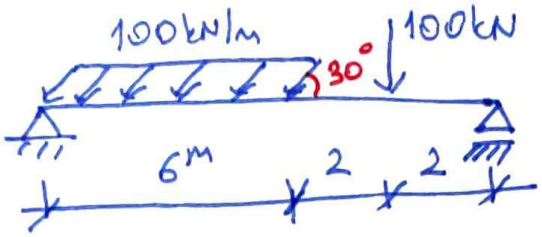

$$X_o = \frac{90}{20}$$

$$X_o = 4,5\,m$$

$$M_{max} = 90.4,5 - 20.\frac{4,5^2}{2}$$

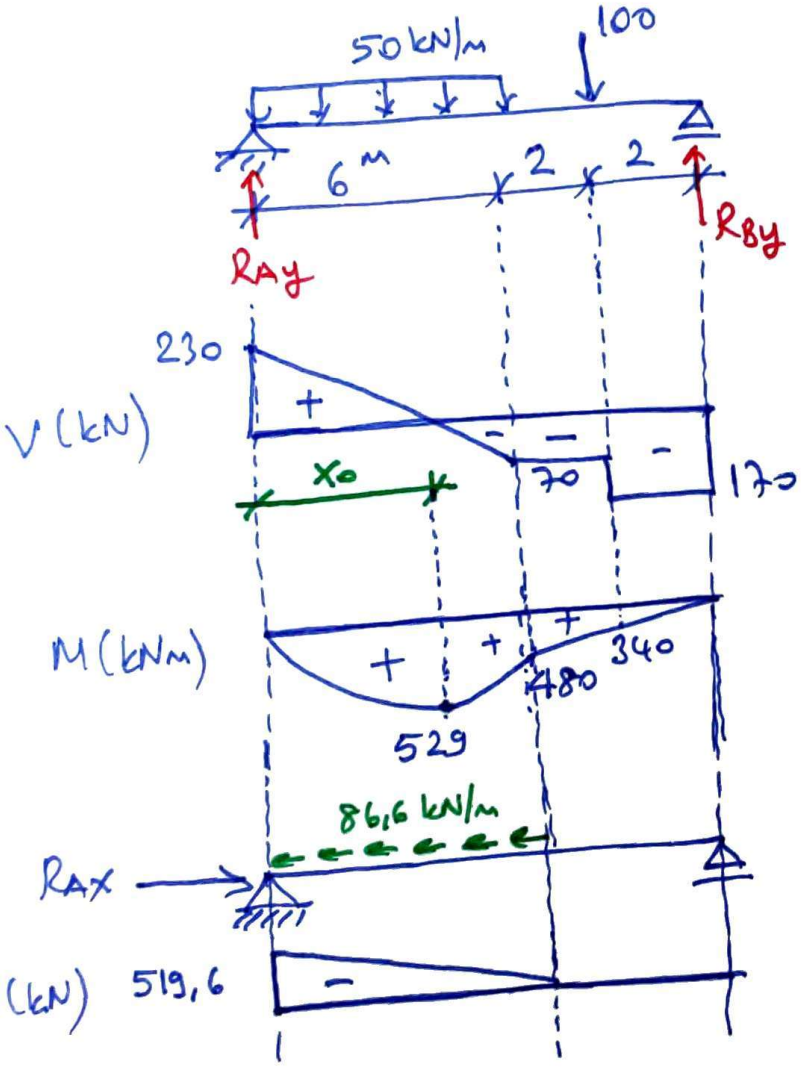
$$M_{max} = 202,5\,kNm$$

15)




$q_x = 100 \cdot \cos 30 = 86{,}6 \text{ kN/m}$

$q_y = 100 \sin 30 = 50 \frac{kN}{m}$




$R_{Ay} = \frac{50 \cdot 6.7 + 100 \cdot 2}{10} = 230 \text{ kN}$

$R_{By} = \frac{50 \cdot 6.3 + 100 \cdot 8}{10} = 170 \text{ kN}$

$x_0 = \frac{230}{50} = 4{,}6 \text{ m}$

$M_{max} = 230 \cdot 4{,}6 - 50 \cdot \frac{4{,}6^2}{2} = 529 \text{ kNm}$

$R_{Ax} = 86{,}6 \cdot 6 = 519{,}6 \text{ kN}$

16)

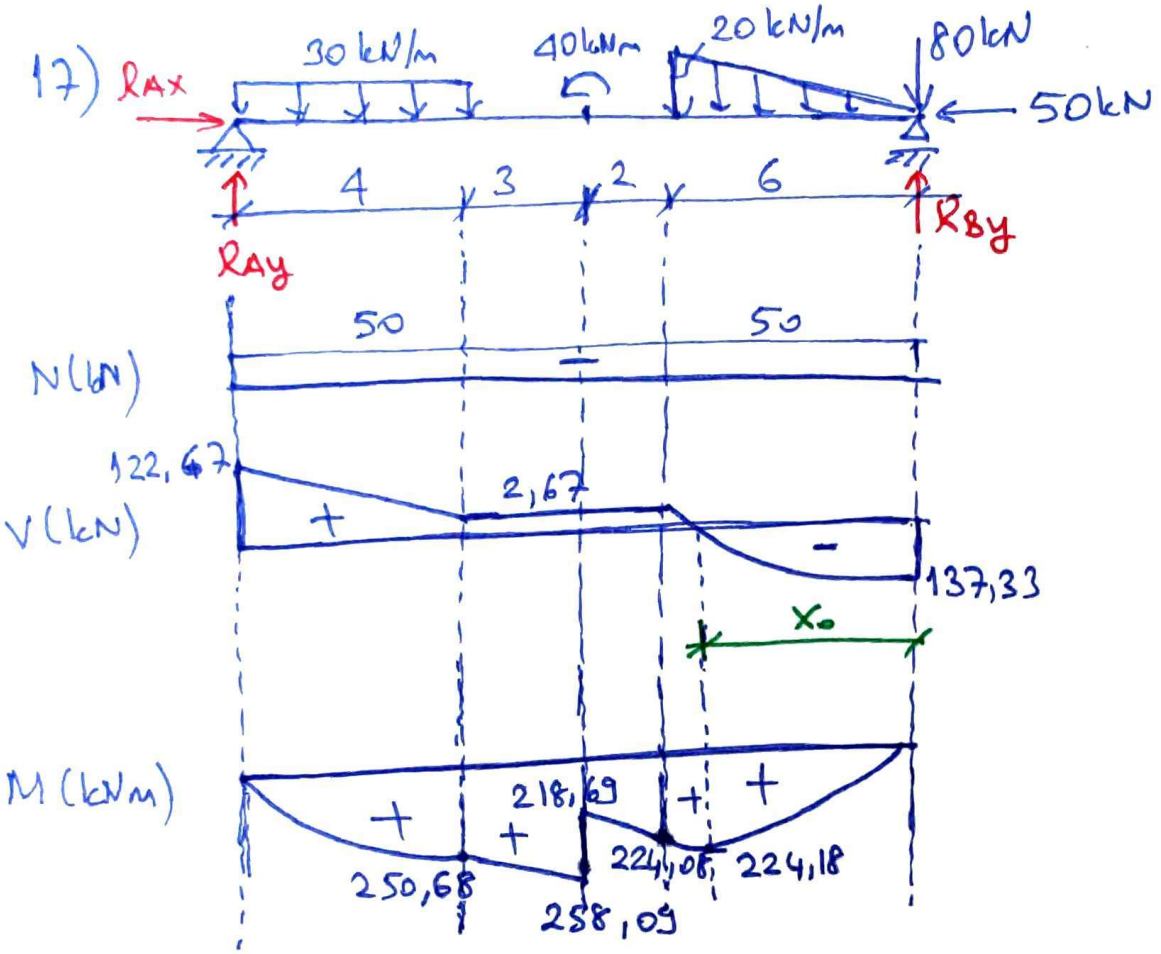



$R_{Ax} = 0$

$R_{Ay} = 0$

$R_{By} = 80 \text{ kN}$

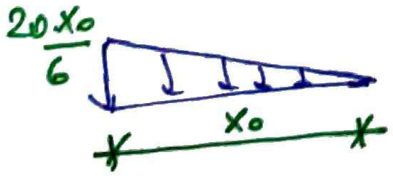
17)




$$R_{Ax} = 50\,kN$$

$$R_{Ay} = \frac{30.4.13 + 40 + 20.6.0{,}5.6.2/3}{15} = 122{,}67\,kN$$

$$R_{By} = \frac{30.4.2 - 40 + 20.3.11 + 80.15}{15} = 137{,}33\,kN$$

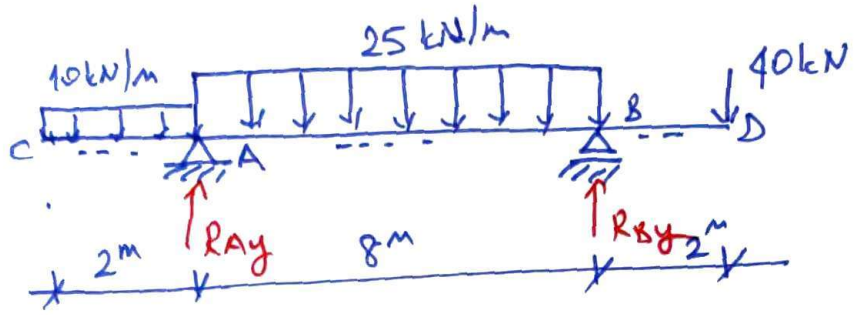



$$20.\frac{x_0}{6}.x_0.\frac{1}{2} = 137{,}33 - 80$$

$$x_0 = 5{,}86\,m$$

$$q(x_0) = \frac{20.x_0}{6} = 19{,}53\,kN/m$$

18)

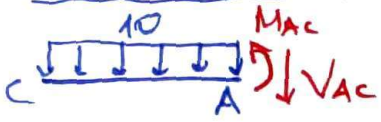



Support Reactions:

$\Sigma M_B = 0 \Rightarrow \quad R_{Ay} = \dfrac{10.2.9 + 25.8.4 - 40.2}{8} = 112,5\ kN$

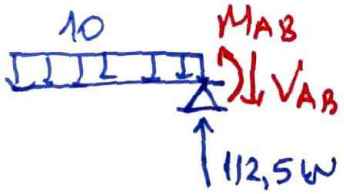
$\Sigma M_A = 0 \Rightarrow \quad R_{By} = \dfrac{40.10 + 25.8.4 - 10.2.1}{8} = 147,5\ kN$

Internal Forces:



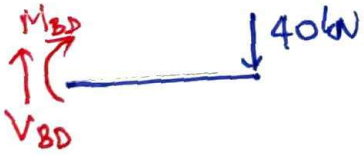

$M_{AC} = -10.2.1 = -20\ kNm$

$V_{AC} = -10.2 = -20$



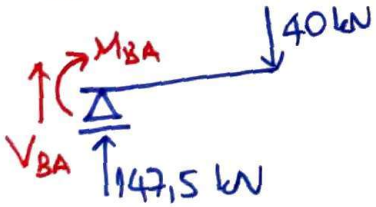

$M_{AB} = -10.2.1 = -20\ kNm$

$V_{AC} = 112,5 - 10.2 = 92,5\ kN$



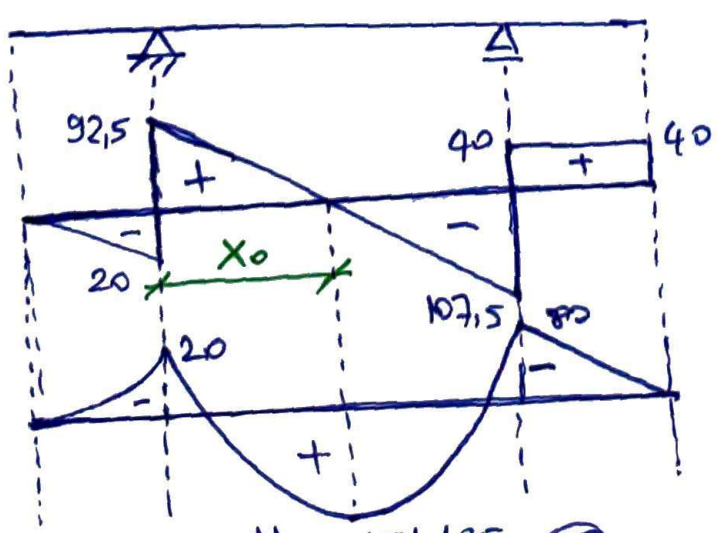

$M_{BD} = -40.2 = -80\ kNm$

$V_{BD} = 40kN$



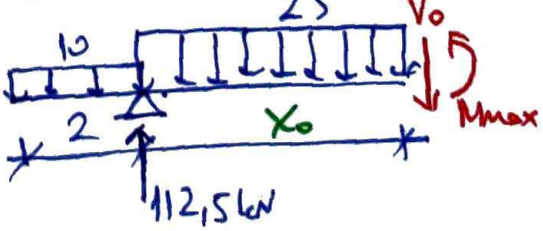

$M_{BA} = -40.2 = -80\ kN$

$V_{BA} = 40 - 147,5 = -107,5\ kN$






$M_{max} = ?$



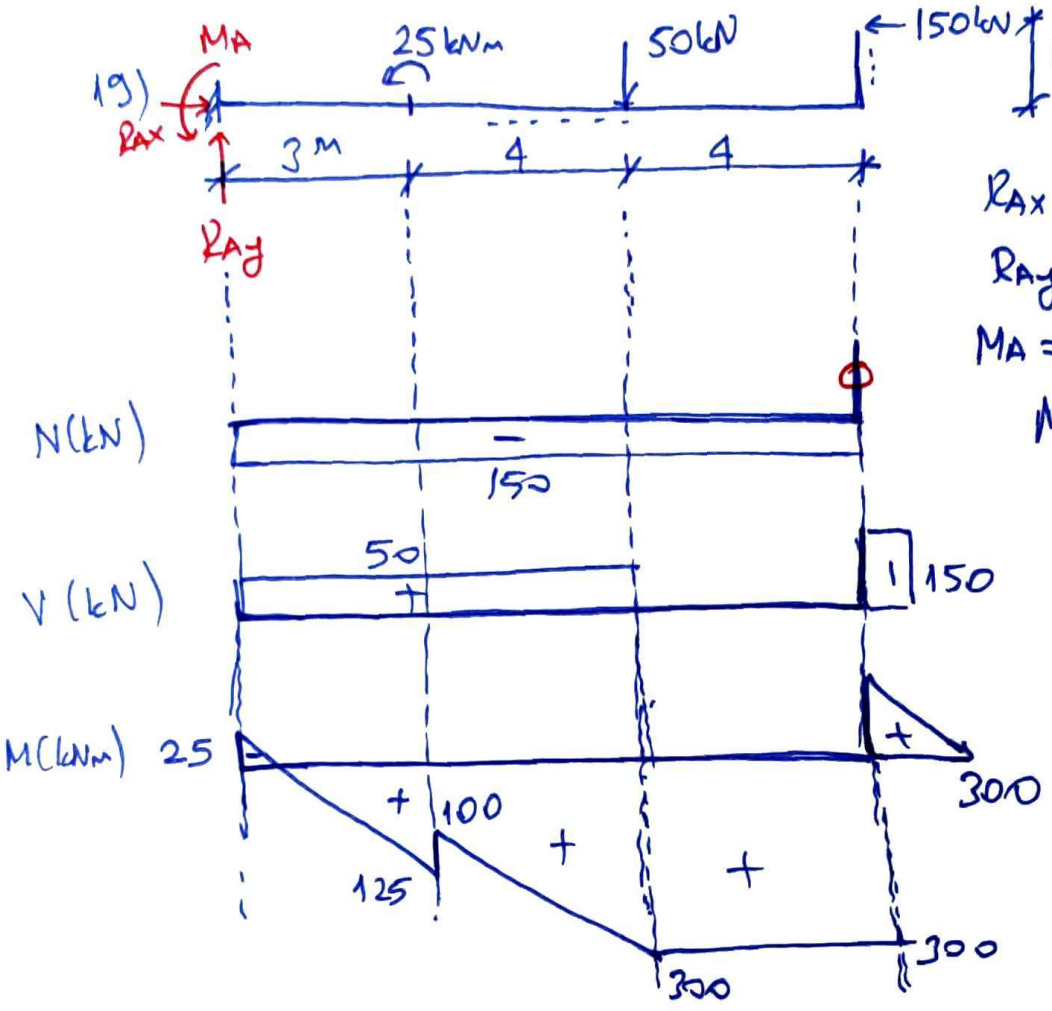

$112,5 = 10.2 + 25.x_0$

$x_0 = 3,7\ m$

$M_{max} = 112,5.3,7 - 10.2.4,7 - 25.\dfrac{3,7^2}{2}$

$M_{max} = 151,125\ kNm$

19)



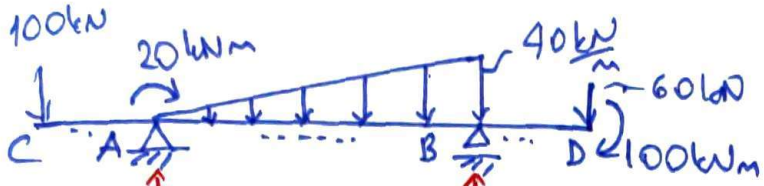

$R_{Ax} = 150\,kN$

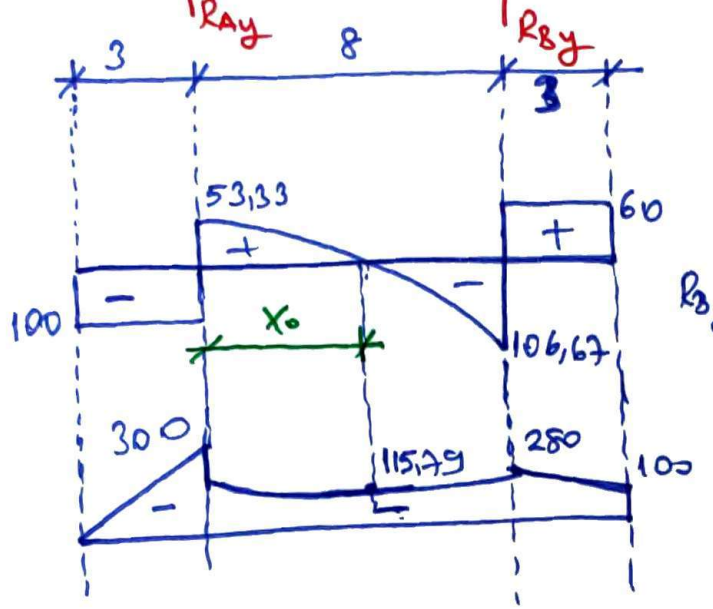
$R_{Ay} = 50\,kN$

$M_A = -25 + 50.7 - 150.2$

$M_A = 25\,kNm$

20)



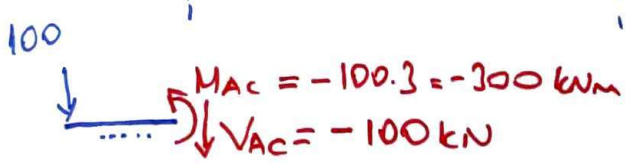



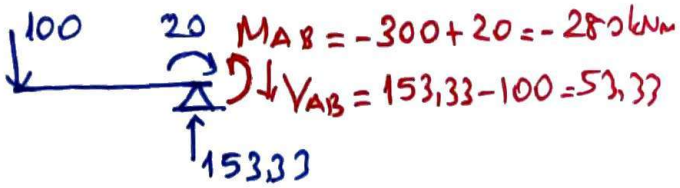

$$R_{Ay} = \frac{100.11 - 20 + \frac{40.8}{2}\frac{8}{3} - 60.3 - 100}{8}$$

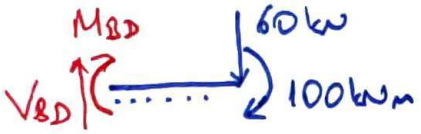
$$R_{Ay} = 153,33\ kN$$

$$R_{By} = \frac{100 + 60.11 + 40.8/2.\frac{2.8}{3} + 20 - 100.3}{8}$$

$$R_{By} = 166,67\ kN$$

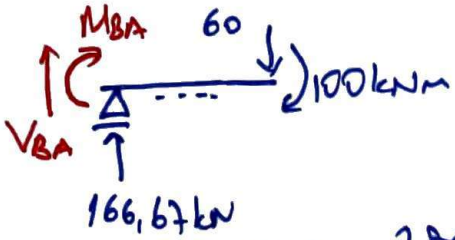


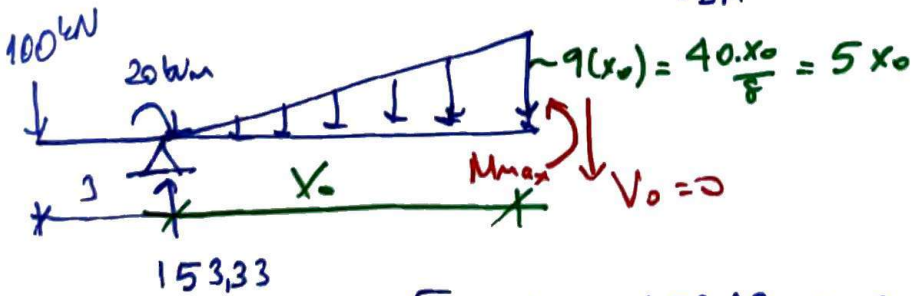

$M_{BD} = -100 - 60.3 = -280\ kNm$

$V_{BD} = 60\ kN$




$M_{BA} = -100 - 60.3 = -280\ kNm$

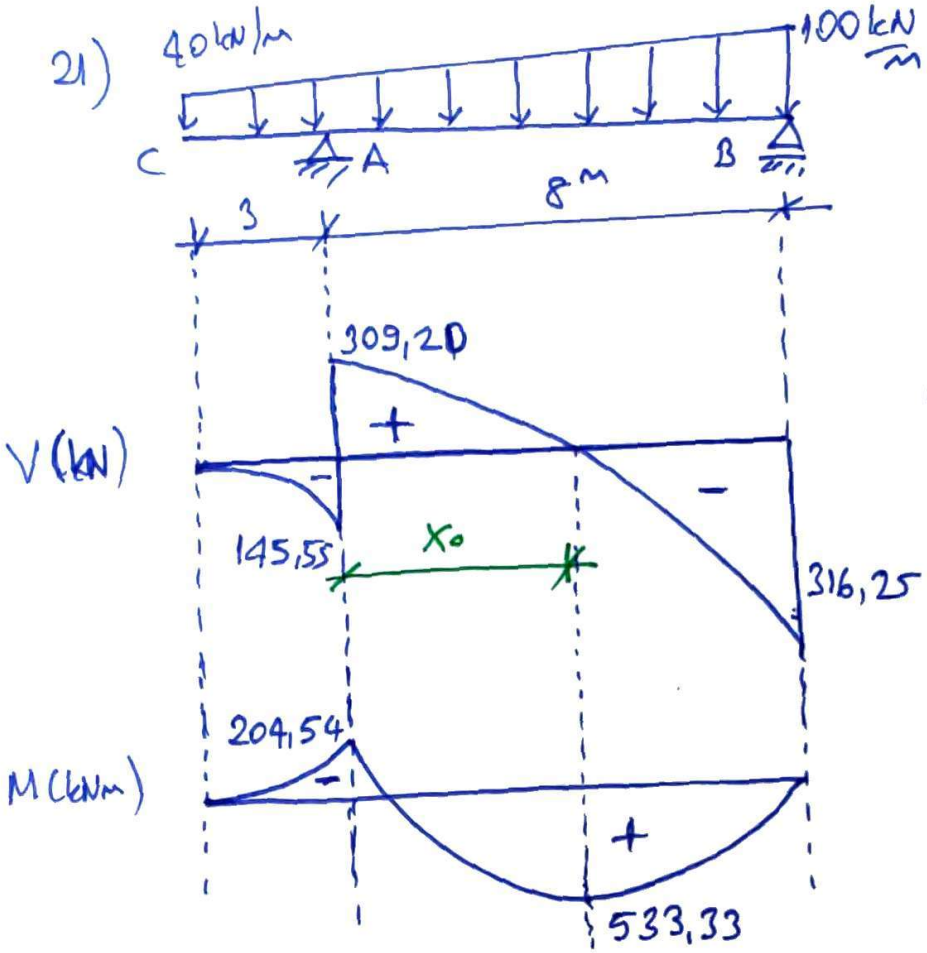
$V_{BA} = 60 - 166,67 = -106,67\ kN$




$$5x_0 \cdot \frac{x_0}{2} = 153,33 - 100 \Rightarrow x_0 = 4,62\ m$$

$$M_{max} = -100.7,62 + 20 + 153,33.4,62 - 5.4,62.\frac{4,62}{2}.\frac{4,62}{3} = 115,79\ kNm$$

21)

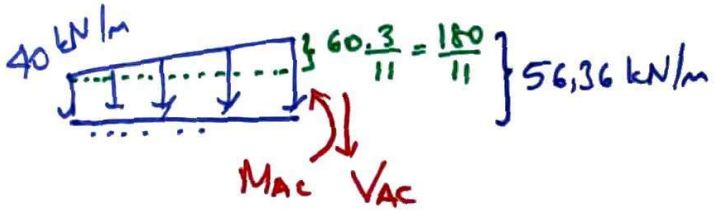



$$R_{Ay} = \frac{40.11.5{,}5 + 60.\frac{11}{2}.\frac{11}{3}}{8}$$

$$= 453{,}75 \text{ kN}$$

$$R_{By} = \frac{40.11.2{,}5 + 60.\frac{11}{2}.(\frac{2}{3}.11+3)}{8}$$

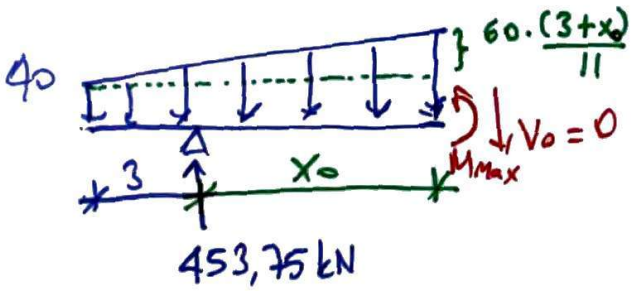
$$= 316{,}25$$




$$V_{AC} = 40.3 - 16{,}36.3.\frac{1}{2}$$

$$V_{AC} = -144{,}55 \text{ kN}$$

$$M_{AC} = -40.3.1{,}5 - 16{,}36.\frac{3}{2}.\frac{3}{3} = -204{,}54$$



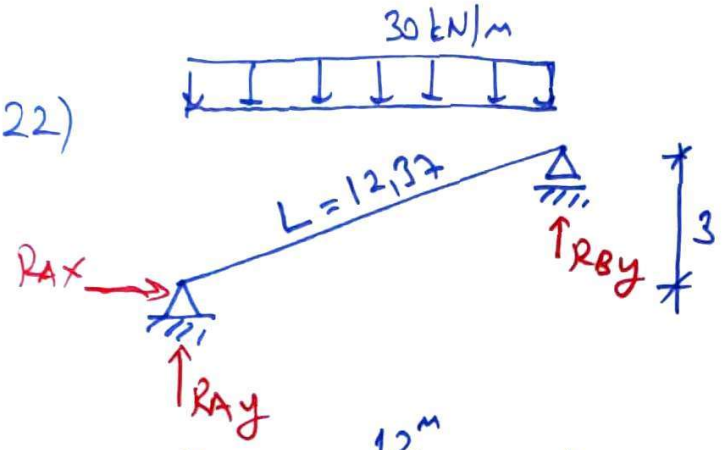

$$453{,}75 = 40.(3+X_0) + \frac{60.(3+X_0)}{11}.(3+X_0).\frac{1}{2}$$

$$X_0 = 4{,}504 \text{ m}$$

$$M_{max} = 435{,}75.4{,}504 - 40.\frac{7{,}504^2}{2} - 60.\frac{7{,}504}{11}.\frac{7{,}504}{2}.\frac{7{,}504}{3}$$

$$M_{max} = 533{,}33 \text{ kNm}$$

22)

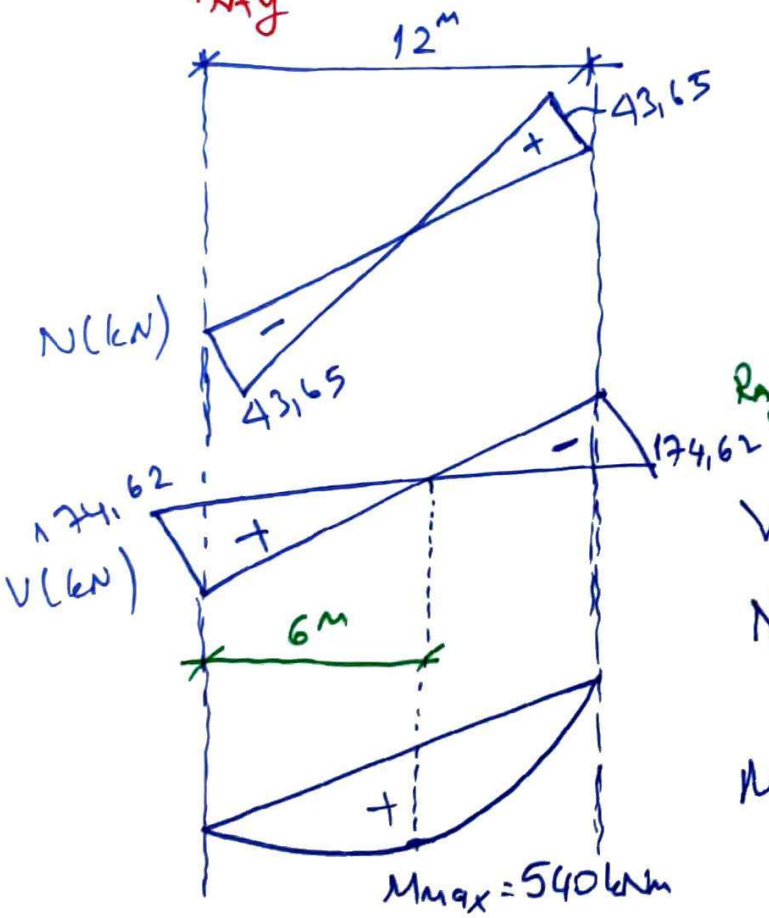


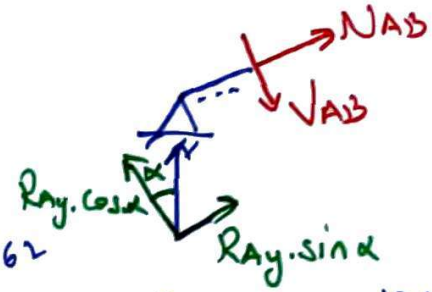

$\sin\alpha = \frac{3}{12,37}$ $\cos\alpha = \frac{12}{12,37}$

$R_{Ax} = 0$

$R_{Ay} = \frac{30.12.6}{12} = 180\,kN$

$R_{By} = \frac{30.12.6}{2} = 180\,kN$




$V_{AB} = R_{Ay}\cdot\cos\alpha = 180\frac{12}{12,37} = 174,62\,kN$

$N_{AB} = -R_{Ay}\cdot\sin\alpha = -180\frac{3}{12,37} = -43,65$

$M_{max} = 180.6 - 30.6.3 = 540\,kNm$

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